THE LEVEL SETS OF THE COUNT OF GEOMETRIC CONNECTED FIBRE COMPONENTS ARE LOCALLY CONSTRUCTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8093-8104 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 8106-8129, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Geometric connected components of a fibre, constructibility of finite-presentation conditions, and base change; the encoding is retained.

Source reference. Original lemma lemma-base-change-fibres-nr-geometrically-connected-components: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7905–7918. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-nr-geom-connected-components-good: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8036–8044. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8129–8129. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 053Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8129–8129. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $n_{X/Y}$ be the function on Y counting the numbers of geometric connected components of fibres of f introduced in Lemma lemma-base-change-fibres-nr-geometrically-connected-components. Assume f of finite presentation. Then the level sets*

$$E_n = \{y \in Y \mid n_{X/Y}(y) = n\}$$

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

of $n_{X/Y}$ are locally constructible in Y.

Proof. Fix n. Let $y \in Y$. We have to show that there exists an open neighbourhood V of y in Y such that $E_n \cap V$ is constructible in V. Thus we may assume that Y is affine. Write $Y = \mathrm{Spec}(A)$ and $A = \mathrm{colim}\, A_i$ as a directed limit of finite type $\mathbf{Z}$-algebras. By Limits, Lemma 01ZM we can find an i and a morphism $f_i : X_i \to \mathrm{Spec}(A_i)$ of finite presentation whose base change to Y recovers f. By Lemma lemma-base-change-fibres-nr-geometrically-connected-components it suffices to prove the lemma for f_i. Thus we reduce to the case where Y is the spectrum of a Noetherian ring.

We will use the criterion of Topology, Lemma 053Z to prove that E_n is constructible in case Y is a Noetherian scheme. To see this let $Z \subset Y$ be an irreducible closed subscheme. We have to show that $E_n \cap Z$ either contains a nonempty open subset or is not dense in Z. Let $\xi \in Z$ be the generic point. Then Lemma lemma-nr-geom-connected-components-good shows that $n_{X/Y}$ is constant in a neighbourhood of ξ in Z. This clearly implies what we want. □